Introduction to Software Engineering

Project Idea & Team Contract

The student team is required to complete the **Project Idea & Team Contract** documentation for the assigned course project, following the attached template.



Software Engineering Department
Faculty of Information and Technology
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Project Idea & Team Contract

Objectives

This document has three objectives:

✔ Present the proposed project concept and scope for teaching-staff review and approval.

✔ Establish a shared agreement on team communication, responsibilities and work standards.

✔ Disclose the team's use of AI tools in accordance with course policy. PA0 is required before PA1, although it is not graded.

1 Team Information

Item	Value
Team ID	T05
Team name	Nexora
Team lead	Nguyễn Đức Hưng
Communication channel	Messenger, Google Meet, Discord

Student ID	Full name	Email	Main role	Backup role
24127040	Dương Trung Hiếu	dthieu2416@clc.fitus.edu.vn	Designer	Frontend Developer; Software Tester
24127139	Lê Thành Trung	lttrung2418@clc.fitus.edu.vn	Software Tester	Business Analyst; Backend Developer
24127174	Nguyễn Đức Hưng	ndhung2431@clc.fitus.edu.vn	Team Lead	Backend Developer
24127519	Bùi Phạm Quang	bpquang2404@clc.fitus.edu.vn	Business Analyst	Full-stack Developer; Software Architect

The primary and backup responsibilities listed above reflect the current team assignments. They may be refined during PA1, subject to team agreement. A backup supports or takes over the corresponding work when necessary.

2 Project Idea Overview

This section introduces the proposed solution, target users, main functions, technical direction and delivery constraints. Detailed requirements and acceptance criteria will be developed in PA2.

2.1 Project Name and One-Sentence Pitch

CourtMate — A web-based sports court booking platform that helps players discover courts, check availability and make reservations, while enabling court owners to manage listings, schedules and bookings.

2.2 Problem and Motivation

Finding a suitable sports court often requires players to contact multiple venue owners by phone or messaging apps to ask about prices, opening hours and available time slots. Information can be scattered or outdated, making it inconvenient to compare options or confirm a reservation. Small venue operators may also manage bookings manually through chat messages, paper notes or spreadsheets. When updates are delayed or records are inconsistent, two players may request the same slot, leading to booking conflicts and extra work for staff.

CourtMate proposes a centralized web platform connecting players with court owners. Players can search courts, view up-to-date court information and available slots, and submit reservations without a lengthy back-and-forth conversation. Owners can publish court details, manage opening schedules and monitor incoming bookings in one place. The core value is a simple, consistent workflow from court discovery to booking and cancellation, with server-side checks to prevent double booking. These are initial problem assumptions for PA0; the team has not claimed to conduct user interviews or quantitative market research.

2.3 Target Users

User group	What they need from the system
Player	Search and compare courts, view court details and available booking times, make or cancel reservations, and manage basic account information.
Court Owner	Publish and update court details and operating schedules, prevent unavailable slots from being booked, and monitor reservations for courts they own.

2.4 Main Features

The proposed scope comprises ten features: eight Must-have features (F01–F08) and two optional Should-have features (F09–F10). Must-have functions form the PA5 baseline. The team will confirm with the TA whether Profile Management should remain separate or be integrated into Authentication.

#	Feature	Short description	Must-have?
1	F01 – Authentication & Authorization	Register, sign in/out, and access functions according to Player or Court Owner roles.	Yes
2	F02 – Browse & Search Courts	Browse listed courts and search by court name, sport type, and area/location.	Yes
3	F03 – Court Details & Availability	View location, supported sport, facilities/description, hourly price, and available dates and slots.	Yes
4	F04 – Court Booking	Select an available booking time and submit a reservation; validate availability and prevent conflicting or overlapping bookings.	Yes
5	F05 – My Bookings	View upcoming and past reservations; cancel an eligible booking under agreed cancellation rules.	Yes
6	F06 – Court & Schedule Management	Owners add, edit, and deactivate courts; configure operating hours and mark unavailable time slots.	Yes
7	F07 – Owner Booking Management	Owners view bookings for their own courts, including dates, time slots, player details, and booking status.	Yes
8	F08 – Profile Management	Users view and edit basic account information; sensitive account actions require authentication.	Yes
9	F09 – Ratings & Reviews	After a completed booking, players may submit a rating and short written review for the court (optional).	No
10	F10 – Promotions	Owners may create simple promotional announcements or discount descriptions; no payment or coupon automation (optional).	No

2.5 Similar Products

1. SanBong.vn
Website: https://sanbong.vn/

SanBong.vn is a Vietnamese sports court booking and management platform. It allows users to search for sports venues, check available time slots, and book courts online. It also provides court owners with features such as booking management, revenue reports, and automated notifications.

2. ALOBO
Website: https://www.alobo.vn/

ALOBO is a sports facility management and booking platform supporting sports such as badminton, football, tennis, and pickleball. Its features include online court reservations, schedule management, customer management, and promotional activities.

How CourtMate Is Different

CourtMate is designed as a focused academic minimum viable product rather than a comprehensive venue-management service. Its priority is a reliable booking lifecycle: court discovery, availability checking, reservation with conflict prevention, cancellation, and essential owner management.

To keep development manageable, CourtMate focuses on court discovery, availability checking, reservations and essential management functions. The booking duration and confirmation workflow have not yet been agreed upon and will be defined during requirements discussions. Online payments, customer relationship management and advanced financial reporting are outside the Must-have scope. Basic promotions may be considered only as a Should-have feature.

The objective is to deliver a clear, dependable and maintainable end-to-end booking workflow within the constraints of a four-member team.

2.6 Platform and Technology Ideas

Platform: Responsive web application for desktop and mobile browsers (no separate mobile app).

Proposed technology stack (to be confirmed during PA1):

Frontend: ReactJS for reusable views, forms, search, calendars and role-based navigation.

Backend: Python with FastAPI, exposing REST APIs for user accounts, court data, availability, and booking operations. Ownership checks and booking validation are enforced on the server.

Database: PostgreSQL for users, court listings, schedules and bookings. Booking operations should use transactions and appropriate constraints or locking to avoid double booking under concurrent requests. Collaboration: GitHub repository, pull requests and GitHub Projects for tasks; Google Docs/Drive for drafts. CI and public hosting are planned for the implementation phase, with final service choices to be confirmed.

2.7 Feasibility, Scope and Open Questions

Feasibility assessment: CourtMate is achievable as a focused academic minimum viable product for four students, although implementing eight Must-have features within the approximately three-week W8–W10 development period presents a significant scheduling challenge. The team will prioritize the core workflow: authentication, court discovery, availability checking, conflict-free reservations, cancellation and owner-side management. Booking duration, time-slot configuration and the confirmation or approval workflow remain open decisions for the team. The proposal excludes online payment integration and anticipates payment at the venue. In PA1, the team will estimate and assign tasks; PA2 will specify detailed business rules and acceptance criteria; subsequent assignments will address design, implementation, testing and deployment.

Deliverability criteria: a working, publicly accessible web application for the Must-have workflows; consistent frontend/backend data and validation; documented setup and test accounts; test evidence for

core and edge cases; and a short demo. If capacity is insufficient, the team should seek TA approval to merge basic profile management into Authentication or simplify the scope before baselining requirements, rather than silently dropping Must-have features.

Out of scope: native Android/iOS apps, online payment processing, real-time chat, AI-based recommendations, QR check-in, recurring reservations, waiting lists, multi-branch administration, complex revenue analytics, and automated coupon redemption. Users are expected to pay at the venue; the platform does not verify or settle payments. The optional Promotions feature, if implemented, will display basic promotional information without automated discounts or online payments.

Key risks and mitigation strategies:

R1 — Booking conflicts: Concurrent requests may reserve the same court and time slot. Mitigation: server-side validation, database transactions or constraints, and concurrency tests.

R2 — Availability errors: Schedule changes and cancellation rules may produce inconsistent slots. Mitigation: define slot boundaries, booking statuses and validation rules early.

R3 — Limited implementation time: The W8–W10 delivery window may cause delays. Mitigation: prioritize Must-have features, assign owners and backups, and review progress weekly.

R4 — Security and authorization: Inadequate access controls may expose user data or permit unauthorized changes. Mitigation: validate inputs, hash passwords and enforce role and ownership checks.

R5 — Integration difficulties: Frontend and backend components may disagree on data formats or workflow. Mitigation: agree on API contracts and test integrations incrementally.

R6 — Deployment issues: Hosting limitations or environment differences may interrupt the final demo. Mitigation: deploy early, document configuration and rehearse the demonstration.

R7 — Insufficient testing: Unchecked edge cases may undermine the booking workflow. Mitigation: prepare positive, negative and concurrency tests and use CI where feasible.

The risk assessment, ownership, probability, impact and contingency actions will be developed in PA1.

Questions for the TA:

1. Our four-member team currently proposes 8 Must-have and 2 Should-have features. Is this scope appropriate for the course timeline? Should profile management be a separate Must-have or part of Authentication?

2. Are there any course constraints on booking duration and the confirmation or approval workflow? Is payment at the venue acceptable without online payment integration?

3. How thoroughly should we demonstrate prevention of concurrent double bookings in PA5?

4. Are free hosting services acceptable for the deployment, and would cold-start delays affect the demonstration?

5. What is the process for obtaining PA0 scope approval before proceeding with PA1?

3 Team Contract

3.1 Goals and Expectations

The team aims to achieve the highest possible course result (10/10) by submitting each PA on time, implementing all agreed Must-have functions, maintaining consistency across requirements, design, implementation and testing, following the engineering process, and delivering a stable deployed system with an effective final demonstration.

Each member is expected to contribute approximately 5–7 hours per week during normal weeks, with additional effort when needed near PA deadlines, implementation, testing, deployment, and the final presentation. Roles define primary responsibility, not exclusive ownership; every member must support other areas when necessary. The team prioritizes correctness, completeness, traceability, evidence of contribution, teamwork, and software quality over unnecessary extra features. Finishing the agreed Must-have scope at high quality takes priority over adding optional features.

3.2 Communication

Messenger is the primary channel for team coordination. Google Meet supports online meetings, while Discord may be used for technical discussion. Urgent issues should be communicated directly to the affected members and Team Lead.

Expected response time to messages: within 12 hours on weekdays for normal project messages. Urgent blockers that may affect another member or a deadline must be reported as soon as possible. If a member is blocked for more than 24 hours, they must report the blocker, what they have tried, and what support is needed.

The planned workflow uses GitHub Projects for task tracking, the project GitHub repository for code and versioned deliverables, and Google Docs/Drive for collaborative drafts. Tool access will be confirmed during PA1. Decisions regarding scope, architecture, ownership and deadlines must be recorded in the task board, meeting minutes or project files—not only in chat.

3.3 Meetings

Weekly status meeting with the TA: according to the official schedule assigned by the TA/course staff.

The team holds two internal meetings per week at 20:00 (Vietnam time, UTC+7). Tuesday is for discussion and planning: PA requirements, decisions, task assignments, reviewers and deadlines. Saturday is for

review and finalization: checking deliverables, resolving feedback and confirming next steps. Any changes to the meeting schedule are agreed upon and announced through the group channel.

Attendance rule: at least 3 of 4 members should attend a normal internal meeting for project decisions. Planned absence should be reported at least 24 hours in advance when possible; unexpected absence must be reported as soon as reasonably possible. An absent member must provide current task status, completed work, blockers, and next steps. Missed meetings are recorded in the weekly status report when appropriate.

The planned weekly status report is prepared by Sunday at 22:00 during weeks 2–10, subject to the official course calendar and team agreement. The Team Lead or backup consolidates updates from task records and meeting minutes, then commits reports to /docs/management/weekly/ in the GitHub repository, unless the TA specifies another procedure.

3.4 Roles and Responsibilities

Team Lead / Project Manager: maintains the project plan and task board, assigns and tracks work, monitors deadlines and blockers, organizes meetings, coordinates weekly status reports, communicates with the TA when necessary, and checks each PA before submission.

Business Analyst: leads problem analysis, scope, requirements, business rules, use cases, and acceptance criteria, and checks consistency of requirements across PAs.

UI/UX Designer / Software Architect: leads interface design and user experience, as well as architecture, C4/UML diagrams, database design and consistency between design and requirements.

Software Developer / Software Tester: supports implementation and integration; plans and executes unit, integration and system testing; tracks defects and assists with continuous integration and test automation.

All members must contribute to implementation, testing, review, integration, and documentation according to their abilities and assigned tasks. Each important role must have a backup member. If a task is reassigned, the previous owner must hand over current files/code, progress, known issues, decisions made, and recommended next steps.

3.5 Work Standards

Internal deadline: each assigned task or document section should be completed at least 48 hours before the official PA deadline whenever possible. A task is considered at risk if the owner reports before the deadline that it may be delayed; it is considered late if the agreed deadline is missed without an approved revised deadline.

Quality: every important document section and pull request must be reviewed by at least one other member. Reviewers check correctness, completeness, consistency with previous PA baselines, traceability, formatting, and whether claims are supported by evidence. Important business logic should not be self-approved by its author. Work that is incomplete, does not run, contradicts the current baseline, lacks required evidence, or contains unverified/fabricated content may be returned to In Progress.

Tools: every significant task must appear on the task board with an owner, due date, status, and evidence/output. Recommended workflow: Backlog → To Do → In Progress → Review → Blocked → Done. A task is Done only when the agreed output is complete, stored/committed in the correct place, reviewed when required, and blocking review comments are resolved. Code changes to main should go through pull requests and at least one review; direct pushes to main should be avoided. Members use their own accounts, write meaningful commits, and never commit passwords, API keys, tokens, or .env secrets.

Scope control: no Must-have feature may be added, removed, or materially changed without team agreement and documentation of the reason and affected requirements/design/tests.

Contribution evidence may include task history, commits, pull requests, reviews, document revision history, meeting notes, and weekly status reports. Equal contribution is not assumed automatically; quality, difficulty, timeliness, responsibility, reviews, support work, testing, documentation, integration, and evidence are considered.

Recognition: members who consistently deliver high-quality work early, help unblock others, perform useful reviews, or take additional responsibility may receive priority in choosing future tasks and explicit recognition in contribution evidence and presentation responsibilities. The team does not impose unofficial course-grade penalties; unresolved contribution issues are documented and escalated to the TA when necessary.

3.6 Use of AI Tools

The team may use generative AI tools such as ChatGPT, Claude, GitHub Copilot, or similar tools for brainstorming, concept explanation, drafting text, code suggestions, debugging ideas, code review, test ideas, and wording improvements. AI output must never be copied blindly. The member who uses AI remains responsible for verifying the output against the project scope, course materials, current requirements/design, and actual implementation.

AI must not be used to fabricate test results, screenshots, commits, contribution evidence, interviews, user feedback, measurements, defects, implementation status, or deployment results. “AI generated it” is not accepted as a reason for incorrect work. Every significant AI use must be recorded in the AI Usage Log for the corresponding PA. Each member reports their own AI usage; the Team Lead checks that the log is complete before submission.

3.7 Decision Making

The team first attempts to reach consensus through discussion. If consensus cannot be reached within a reasonable time, the decision is made by majority vote. Because the team has four members, a 2–2 tie is possible; in that case, the Team Lead makes the final decision after considering the recommendation of the member responsible for that area (e.g., Business Analyst for requirements, Software Architect for design, Software Developer for implementation, Software Tester for testing). Major decisions affecting Must-have scope, architecture, requirements, deadlines, or responsibilities must be recorded.

3.8 Handling Conflicts and Missed Work

Step 1: talk directly with the member, identify the cause, agree on a concrete fix and a new deadline, and determine what support is needed. Reporting a risk/blocker before the deadline is encouraged and is treated differently from silently missing a deadline.

Step 2: if the issue continues or affects project progress, discuss it in a team meeting, record it in the weekly status report/meeting notes, agree on corrective action, and increase monitoring when necessary.

Step 3: if critical work is at risk, part or all of the task may be reassigned to a backup member with a proper handover. Repeated missed deadlines, non-communication, repeated absence, or unusable work must be supported by evidence such as tasks, commits, pull requests, reviews, revision history, and weekly reports.

Step 4: if the problem cannot be resolved internally or serious/persistent non-contribution continues, inform the TA and provide the available evidence. Individual contributions should reflect actual documented work rather than automatically being divided equally.

3.9 Signatures

By signing, each member confirms that they have read, discussed, understood, and agreed to follow this contract. The contract may be reviewed by the team when circumstances change, but material changes must be discussed, agreed upon, and documented.

Student ID	Full name	Signature	Date
24127040	Dương Trung Hiếu	Hiếu	10/10/2026
24127139	Lê Thành Trung	Trung	10/10/2026
24127174	Nguyễn Đức Hưng	Hưng	10/10/2026
24127519	Bùi Phạm Quang	Quang	10/10/2026

4 AI Usage Log

This log records significant uses of generative AI during PA0, the tasks supported and how outputs were reviewed or adapted. Members are responsible for disclosing their own AI usage and checking submitted work in accordance with the course policy.

#	Member	Tool	Task / purpose	How the output was verified and changed
1	Lê Thành Trung	ChatGPT	Brainstorm CourtMate features and risks	Reviewed with team and made a final decision
2	Nguyễn Đức Hưng	ChatGPT	Brainstorming project ideas, defining features and drafting PA0 content	Reviewed against PA0 requirements; revised the scope and wording
3	Bùi Phạm Quang	Gemini	Recommendations for main features and technology stack selection	Compared options against project constraints; selected a provisional technology stack